

The Sankey diagram illustrates the flow of data from various sources to RNAseq and microarray datasets. The sources are categorized into primary, circulating, and cell line. The primary sources include bronchoalveolar lavage (BAL) (1), blood (5), CD14+ MDMs MF1 (2), CD14+ MDMs MF2 (1), PBMC (4), PBMC macrophages (1), dendritic cells (3), macrophages (3), monocytes (1), MDMs (3), lymphnode (1), HepG2 (2), and THP1 (1). The circulating sources include MTB (25), EPTB (1), PTB (1), and LNTB (1). The cell line sources include HepG2 (2) and THP1 (1). The data flows into two main categories: RNAseq (22) and microarray (6). The RNAseq category includes 0h (1), 24h (2), 2h (1), 48h (2), 4h (1), not applicable (22), and not indicated (21). The microarray category includes GPL10558 (5) and GPL16250 (1).

Source	Count	Destination	Count
bronchoalveolar lavage (BAL) (1)	1	primary	1
blood (5)	5	primary	5
CD14+ MDMs MF1 (2)	2	primary	2
CD14+ MDMs MF2 (1)	1	primary	1
PBMC (4)	4	primary	4
PBMC macrophages (1)	1	primary	1
dendritic cells (3)	3	primary	3
macrophages (3)	3	primary	3
monocytes (1)	1	primary	1
MDMs (3)	3	primary	3
lymphnode (1)	1	primary	1
HepG2 (2)	2	cell line	2
THP1 (1)	1	cell line	1
MTB (25)	25	circulating	25
EPTB (1)	1	circulating	1
PTB (1)	1	circulating	1
LNTB (1)	1	circulating	1
other (4)	4	circulating	4
primary (25)	25	RNAseq	22
circulating (24)	24	RNAseq	21
cell line (3)	3	microarray	6

The diagram illustrates the drug prediction pipeline. It starts with **Individual disease signatures** (represented by multiple red bars). These are processed through a **Weighted average method** (step 2) to create an **Aggregated disease signature** (a single red bar). This aggregated signature is then combined with **Disease-drug connectivity methods** (step 1) to produce **Drug signatures ranked by reversal scores**. These scores are used to generate **Ranked lists of drug predictions** for **six chosen methods**. The final output is **Drug signatures** (represented by multiple blue bars).

Enrichment-based methods include:

- CMAP 1.0:
 - Connectivity score (**CS**)
- LINCS scores:
 - Weighted Connectivity Score (**WCS**)
 - Normalized Connectivity Score (**NCS**)
 - Tau score

Correlation-based methods include:

- Extreme Pearson (**XCor**)
- Extreme Spearman (**XSpe**)

Ranked lists of drug predictions example:

Drug sig (drug_cell)	Score
C_a	-0.9
F_i	-0.85
...	
A_o	0.95

C.

Microarray individual signatures (n = 6)

RNAseq individual signatures (n = 22)

Microarray aggregated signatures

RNAseq aggregated signatures

Drug signatures ranked by reversal scores

Six chosen methods

Drugs consistently reversing TB disease signatures (≥ 3 signatures; top 20% reversal)

Drug signatures with the top 10% most reversal scores

Drug sig Score (drug_cell)

Drug sig Score (drug)

Selecting drugs consistently prioritized across methods

Partial rank aggregation (BiG approach) based on method categories

Lists of drug signatures ranked by rank z-scores

union

intersection

High-confidence drug predictions
(drugs with both positive individual and aggregated signatures' rank mean z-scores)